

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
6	(a)	(i)		Break down of (large /insoluble) to {small molecules / soluble} (1) so that they can be <u>absorbed</u> (1)	2			2		
		(ii)		Contraction (and relaxation) of muscles in gut wall (1) Push food along (1)	2			2		
	(b)	(i)		Increase (in temp 20-40) and decreases (40-60) (1) (Optimum at) 40 – fastest rate (1) kinetic energy increasing/ more successful collisions between 20-40 (1) enzyme is denature / active site changes shape 40-60 (1)	2	2		4		4
		(ii)		pH / partial denaturation / different enzyme	1			1		1
		(iii)		boiled <u>and</u> cooled enzyme (1)	1			1		1
		(iv)		Above 65°C Bacterial amylase has <u>higher</u> rate / tolerance at <u>higher</u> temperatures / Optimum at a <u>higher</u> temperature / Less active at lower temperatures (1) Denatures at a <u>higher</u> temperature (1) Allow correct comparison of data for first marking point (1)		2		2		2
				Question 6 totals	8	4		12	0	8